

Laboratory 5: Where do errors come from?

1. Round-off errors.

Using the command `evalf` in Maple, evaluate $25^{1/8}$. Then, raise the result to the power of eight. What do you expect to obtain?

Iterations of the logistic map: from order to chaos.

Theory. We define the sequence $(x_k)_{k \geq 0}$ of iterations of a function f starting with a given $x_0 \in \mathbb{R}$ as

$$x_{k+1} = f(x_k), \text{ for all } k \geq 0,$$

i.e. $x_1 = f(x_0)$, $x_2 = f(x_1)$, $x_3 = f(x_2)$,

A fixed point of a function f is a number $\eta^* \in \mathbb{R}$ such that $f(\eta^*) = \eta^*$. Thus, in order to find the fixed points of f one has to solve the equation

$$f(x) = x.$$

Note that η^* is a fixed point of the function f if and only if the sequence of iterations of f starting with η^* is constant, i.e. $x_k = \eta^*$ for all $k \geq 0$.

2. Let $f1(x) = x(1 - x)$.

(a) Find the fixed points of $f1$.

(b) Plot the graphs of $f1$ and the line $y = x$ for $x \in [0, 1]$.

Note that the two graphs intersect in $(0, 0)$ and (η^*, η^*) where η^* is another fixed point of $f1$ (if it exists).

Note also that $f1(x) \in [0, 1]$ for all $x \in [0, 1]$.

(b) Compute the first 200 iterations of $f1$ starting with $x_0 = 0.5$ and represent them using `pointplot` in Maple. Interpret the results.

(c) Repeat (b) for the initial values $x_0 = 0$ and, respectively, $x_0 = 0.7$. You can try also with other initial values if you want.

3. Repeat exercise 2 for the function $f2(x) = 2x(1 - x)$.

4. Repeat exercise 2 for the function $f4(x) = 4x(1 - x)$.

Definition of the logistic map.

$$\lambda \in [1, 4], \quad f_\lambda(x) = \lambda x(1 - x), \quad x \in [0, 1].$$

Possible behaviors of the sequence of iterations of the logistic map.

- (i) it is constant (this happens only if the starting point is a fixed point of f_λ);
- (ii) it converges monotonically to a certain fixed point of f_λ ;
- (iii) it can be partitioned into two subsequences that converge to distinct values;
- (iv) it can be partitioned into four subsequences that converge to distinct values;
- (v) it has no pattern, it is chaotic.

5. Repeat exercise 2 for the logistic map for the following specific values of the parameter

$$\lambda = 3.1, \quad \lambda = 3.5, \quad \lambda = 3.55, \quad \lambda = 3.6, \quad \lambda = 3.8.$$

It is important to consider them all!

6. Sensitivity with respect to small errors is a feature of chaos.

(a) Let $x_0 = 0.67$ and $y_0 = 0.67001$. Let

$$x_{k+1} = 4x_k(1 - x_k), \quad y_{k+1} = 4y_k(1 - y_k).$$

Compute, in the same **for**, the first 40 terms of $(x_k)_{k \geq 1}$ and $(y_k)_{k \geq 1}$. Compare x_k and y_k .

(b) Let $x_0 = 0.67$, $y_0 = 0.67$ and $z_0 = 0.67$. Let

$$x_{k+1} = 4x_k(1 - x_k), \quad y_{k+1} = 4y_k - 4y_k^2, \quad z_{k+1} = 4z_k - 4z_k z_k.$$

Compute, in the same **for**, the first 40 terms of $(x_k)_{k \geq 1}$, $(y_k)_{k \geq 1}$ and $(z_k)_{k \geq 1}$.

What do you expect and what you obtain?

From iteration to approximation: numerical methods for differential equations

Theory. We consider the IVP

$$x' = f(t, x), \quad t \in [t_0, t^*], \quad x(t_0) = x_0.$$

Let $n \geq 1$ (the number of steps). The Euler method formula with constant step size $h = (t^* - t_0)/n$ is

$$x_{k+1} = x_k + h f(t_k, x_k), \quad t_{k+1} = t_k + h, \quad k = \overline{0, n-1}.$$

The improved Euler method formula with constant step size $h = (x^* - x_0)/n$ is

$$x_{k+1} = x_k + \frac{h}{2} f(t_k, x_k) + \frac{h}{2} f(t_k + h, x_k + h f(t_k, x_k)), \quad t_{k+1} = t_k + h, \quad k = \overline{0, n-1}.$$

The output is a time sequence, equally distributed in the interval $[t_0, t^*]$,

$$t_1, \quad t_2, \quad \dots \quad t_n = t^*$$

and a sequence of values

$$x_1, \quad x_2, \quad \dots \quad x_n.$$

If φ is the exact solution of the given IVP, then the value x_k is an approximation of $\varphi(t_k)$, i.e.

$$x_k \approx \varphi(t_k) \quad \text{with the error} \quad \varepsilon_k = |\varphi(t_k) - x_k|.$$

7. We consider the IVP $x' = 2tx$, $t \in [0, 1]$, $x(0) = 1$.

- (a) For the step size $h = 0.1$, how many steps n are necessary to cover the interval $[0, 1]$? What are the values of $t_1, \dots, t_n$?
- (b) Apply the Euler's method with step size $h = 0.1$ in the interval $[0, 1]$.
- (c) Apply the improved Euler's method with step size $h = 0.1$ in the interval $[0, 1]$.
- (d) Find the exact solution of the IVP. Denote it by φ .
- (e) For each method, find the error of approximating $\varphi(0.5)$ and, respectively, $\varphi(1)$.

It seems that the errors increase with the number of steps. It seems that the errors are smaller for the improved Euler's method.

8. We consider the IVP $x' = t^2 + x^2$, $t \in [0, 2]$, $x(0) = 0$.

- (a) For the step size $h = 0.1$, how many steps n are necessary to cover the interval $[0, 2]$? What are the values of $t_1, \dots, t_n$?
- (b) Apply the Euler's method with step size $h = 0.1$ in the interval $[0, 1]$.
- (c) Apply the improved Euler's method with step size $h = 0.1$ in the interval $[0, 1]$.
- (d) Use *DEplot* to represent the direction field of the differential equation and the graph of the solution of the IVP. Note that this is the graph of an approximate solution, which is found with a Runge-Kutta type numerical method.

For what reason are these results so different near 2? What do you think? It seems that on the intervals $[0, 1]$, or even $[0, 1.5]$ the approximations are quite good!

9. What is a stiff equation? We consider the IVP $x' = -250x$, $x(0) = 1$.

- (a) Find the exact solution of the IVP.
- (b) Apply the Euler's method with step size $h = 0.1$ in the interval $[0, 1]$. Compute the error of approximation at each step. Huge error!
- (c) Apply the improved Euler's method with step size $h = 0.1$ in the interval $[0, 1]$. Compute the error of approximation at each step. Huge error!